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Gesture-Controlled Excel Management System

Multimodal Interaction - 2ºAssignment



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1. Introduction

This document presents the design and implementation of a **gesture-controlled system for managing and analysing data in Microsoft Excel**. The system enables users to interact with spreadsheets through **natural hand gestures**, eliminating the need for traditional input devices such as the mouse and keyboard.

The interaction is based on a **Kinect sensor**, which captures user movements and recognises predefined gestures. These gestures are mapped to Excel actions such as calculating averages, navigating between cells and worksheets, inserting columns, applying conditional formatting, generating charts, deleting cell values, and closing the application.

The system follows a **multimodal architecture**, where gesture recognition is handled by the Kinect and Visual Gesture Builder, gesture events are processed by the MMI Fusion Engine, and Excel manipulation is performed through a C# backend using **Microsoft Excel Interop**. This separation ensures reliable command execution and real-time feedback.

The main goal of this project is to **increase accessibility, reduce cognitive load, and minimise user errors** when working with Excel, particularly for users with limited digital skills or in environments where silent, hands-free interaction is required (e.g., shared offices or classrooms).

Overall, this project demonstrates how gesture-based interaction can make spreadsheet manipulation more intuitive, efficient, and inclusive.

2. Functionalities and Implemented Gestures

This section describes each implemented gesture, its purpose, and the corresponding Excel functionality, fully aligned with the system demonstrated in the presentation (slides 9–17).

2.1 Navigation and Interaction Gestures

2.1.1 Zoom In / Zoom Out

Function: Adjusts the spreadsheet zoom level to improve data readability without manual navigation.

Gesture: Two distinct hand gestures representing zoom-in and zoom-out motions.

Excel Action: Increases or decreases the worksheet zoom level, allowing the user to focus on specific data areas.

2.1.2 Swipe Up / Down

Function: Enables fast vertical navigation across spreadsheet cells.

Gesture: Vertical swipe movement of the hand.

Excel Action: Moves the active cell up or down, supporting efficient exploration of large datasets.

2.1.3 Swipe Left / Right

Function: Enables fast horizontal navigation across spreadsheet cells.

Gesture: Horizontal swipe movement of the hand.

Excel Action: Moves the active cell left or right.

2.2 Excel Manipulation

2.2.1 Calculate Average

Function: Automates the calculation of averages, eliminating the need to write formulas manually.

Gesture: Predefined hand gesture associated with average calculation.

Excel Action: Calculates the average of selected cells or predefined ranges.

2.2.2 Highlight Approved / Failed Students

Function: Supports visual analysis of student performance.

Gesture: Hand gesture mapped to conditional formatting.

Excel Action: Applies conditional formatting to highlight passing students and failing students using predefined thresholds.

2.2.3 Insert Column

Function: Enables simple and efficient data organisation.

Gesture: Gesture representing insertion action.

Excel Action: Inserts a new column at the appropriate position in the spreadsheet.

2.2.4 Delete Cell Value

Function: Allows error correction without direct keyboard or mouse interaction.

Gesture: Deletion gesture.

Excel Action: Clears the content of the currently selected cell without affecting the spreadsheet structure.

2.2.5 Close Excel

Function: Closes the Excel application safely.

Gesture: Closing gesture.

Excel Action: Saves changes and closes Excel without displaying confirmation pop-ups.

3. Gesture Recognition and Training

Gesture recognition is performed using the **Kinect Xbox sensor** combined with **Visual Gesture Builder**. Each gesture was trained using multiple annotated examples to ensure reliable recognition and robustness. The number of annotated examples varies per gesture, with confidence thresholds set to 0.6 to balance accuracy and responsiveness.

Special attention was given to designing **distinct and non-conflicting gestures**, reducing false positives and missed detections. This training process ensures consistent behaviour across different users and movement styles.

4. System Architecture

The system follows a modular multimodal architecture, as illustrated in slide 8:

- **Input Layer:** Kinect sensor captures user movements.
- **Gesture Recognition Layer:** Visual Gesture Builder identifies predefined gestures.
- **Fusion Layer:** The MMI Fusion Engine combines gesture events and maps them to application commands.
- **Application Layer:** A C# backend uses Microsoft Excel Interop to execute spreadsheet actions and provide visual feedback.

This separation of responsibilities ensures real-time responsiveness, reliability, and extensibility.

5. System Usage Instructions

1. Start the Kinect sensor and ensure correct positioning.
2. Launch the gesture recognition module and the MMI Fusion Engine.
3. Run the Excel backend application.
4. Open the target Excel file.
5. Perform supported gestures to interact with the spreadsheet.
6. Observe immediate visual feedback in Excel.

6. Conclusion

The Gesture-Controlled Excel Management System demonstrates how gesture-based interaction can make Excel more intuitive, accessible, and efficient. By replacing complex menus and formula-based interactions with natural gestures, the system reduces cognitive load and minimises user errors.

The project successfully integrates gesture recognition, multimodal fusion, and Excel automation, enabling hands-free interaction for common spreadsheet tasks. Future work includes expanding the gesture set and combining gesture interaction with voice-based modalities.

The Gesture-Controlled Excel Management System demonstrates how gesture-based interaction can make Excel more intuitive, accessible, and efficient. By reducing dependence on menus, formulas, and traditional input devices, the system lowers cognitive load and minimises user errors.

The project successfully integrates gesture recognition with Excel automation, enabling hands-free interaction for common spreadsheet tasks. Future work includes adding more gestures and combining gesture interaction with other modalities.

7. References

- Microsoft Excel Interop Documentation: <https://learn.microsoft.com/en-us/dotnet/csharp/advanced-topics/interop/how-to-access-office-interop-objects>
- Kinect and Visual Gesture Builder documentation
- Lecture demonstration materials